Binary field multiply: two Modal x86-64 CPU panels

Status: two completed field-level panels; descriptive timing only. The full report and raw receipts are in research/ecc_arithmetic_20261013/reports/modal-20261013/REPORT.md.

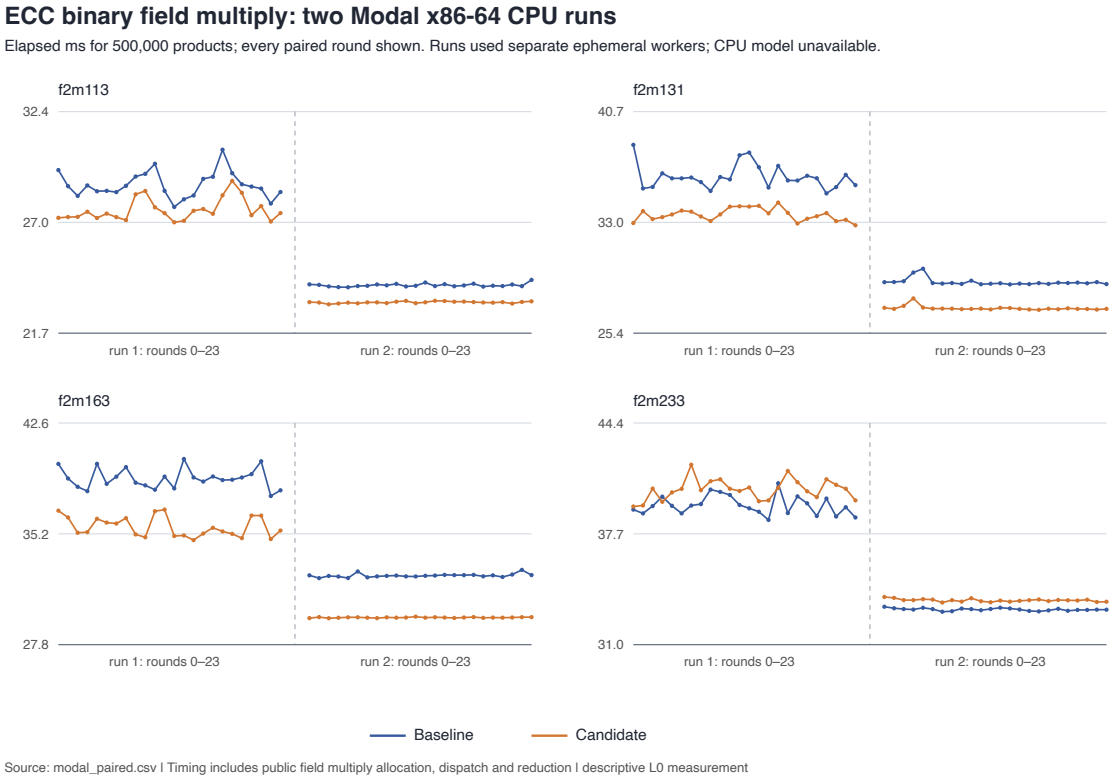
Execution and correctness

The same frozen six Rust files and a corrected launcher ran twice on separate Modal workers. Baseline source SHA-256: 2bc2a1f55887ef7db1fdb1ce5a6cd05c86b7c75e9eb8ade36497ff5686b37fd3. Candidate source SHA-256: 8b32c5467ee8c87752b9cb86594fb742aa9053fd5918172686c4270c9e936364. Each worker requested two vCPUs, 2,048 MiB and a 1,200-second timeout. It reported Linux/gVisor x86-64, PCLMULQDQ and VPCLMULQDQ, but no CPU model or reserved-core receipt, remote timestamps, or per-phase resource data. Each passed 34 tests, then completed 24 alternating AB/BA pairs for four fields at 500,000 public multiplies per arm per pair. All 384 arm rows are retained; all paired checksums match. The first launcher attempt failed at a remote source path and has no timed rows.

Descriptive timing

Median ms for 500,000 public field multiplies, baseline / candidate:

Field	Run 1	Run 2	Product count
GF(2^113)	28.767 / 27.471	23.996 / 23.200	4 ⟶ 3
GF(2^131)	36.079 / 33.580	28.825 / 27.043	9 ⟶ 6
GF(2^163)	38.869 / 35.389	32.390 / 29.583	9 ⟶ 6
GF(2^233)	39.378 / 40.410	33.142 / 33.702	unchanged control



Absolute times shifted markedly between workers, including the unchanged control. The model, placement, frequency and thermal state are unreported; there is no A/A or production-library comparison. This does not pass the controlled wall-time gate and establishes no qualified speedup or worldwide rank. Timing covers public field multiply allocation, dispatch and reduction, not point/scalar arithmetic or an ECC2K walk.